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Task-specific reach-to-grasp training after stroke: Development and description of a home-based intervention

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Abstract

Objective: To describe and justify the development of a home-based, task-specific upper limb training intervention to improve reach-to-grasp after stroke and pilot it for feasibility and acceptability prior to a randomized controlled trial.

Intervention description: The intervention is based on intensive practice of whole reach-to-grasp tasks and part-practice of essential reach-to-grasp components. A 'pilot' manual of activities covering the domains of self-care, leisure and productivity was developed for the feasibility study. The intervention comprises 14 hours of therapist-delivered sessions over six weeks, with additional self-practice recommended for 42 hours (i.e. one hour every day). As part of a feasibility randomized controlled trial, 24 people with a wide range of upper limb impairment after stroke experienced the intervention to test adherence and acceptability. The median number of repetitions in one-hour therapist-delivered sessions was 157 (interquartile range IQR 96–211). The amount of self-practice was poorly documented. Where recorded, the median amount of practice was 30 minutes (interquartile range 22–45) per day. Findings demonstrated that the majority of participants found the intensity, content and level of difficulty of the intervention acceptable, and the programme to be beneficial. Comments on the content and presentation of the self-practice material were incorporated in a revised 'final' intervention manual.

Discussion: A comprehensive training intervention to improve reach-to-grasp for people living at home after stroke has been described in accordance with the Template for Intervention Description and Replication (TIDieR) reporting guidelines. The intervention has been piloted, and found to be acceptable and feasible in the home setting.

Trial registration: ISRCTN56716589

Keywords

Stroke, hand, arm, upper limb, physical therapy, occupational therapy, rehabilitation, home, reach, grasp, repetitive, task-specific training, practice

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Introduction

Task-specific training is recommended internationally in stroke rehabilitation guidelines[1–4] and involves intensive practice of actions or functional tasks.[5] A Cochrane overview of systematic reviews concluded that there was moderate quality evidence that at least 20 additional hours of repetitive task training was effective for improving upper limb function after stroke, but that sufficiently powered, high-quality randomized controlled trials were required to strengthen the evidence.[6] Task-specific upper limb training specifically focusing on reach-to-grasp movements would appear to be particularly relevant, as stroke survivors consulted about their goals for upper limb therapy programmes prioritised activities involving reach-to-grasp.[7,8] Reach-to-grasp movements are essential for everyday functions, such as retrieving objects, e.g. clothes, food and drink, and are used more frequently than other upper limb movements, such as gesturing, stabilising objects or for postural support.[9]

With the implementation of early-supported discharge services, upper limb rehabilitation after stroke is increasingly being provided at home. Home-based training is likely to be influenced by environmental and psychosocial factors and currently there is insufficient evidence for the effectiveness of home-based therapy for improving functional recovery of the upper limb after stroke.[10] Therefore, interventions involving high-intensity task-specific training focusing on reach-to-grasp activities need to be developed for the home setting.

Developing feasible, evidence-based, meaningful therapeutic interventions is a key aim of neurological rehabilitation research.[11] However, only 39% of non-pharmacological interventions evaluated in randomized trials are adequately described,[12] meaning clinicians do not have sufficient information to deliver tested interventions to patients.[13]

The intervention was developed in the context of the UK Medical Research Council (MRC) framework for developing and evaluating complex interventions, which describes the need to identify evidence, model the intervention for delivery and test feasibility, prior to testing effectiveness in a definitive evaluation.[14,15] The aim of this article is to explain the development process, describe and justify the essential components of this intervention and report its feasibility and acceptability among stroke survivors and therapists. The work was carried out as part of a feasibility randomized controlled trial of a home-based task-specific reach-to-grasp training intervention for people after stroke (Trial registration: ISRCTN56716589).

Intervention description

The 'Template for Intervention Description and Replication' (TIDieR) checklist[16] has been used to structure the description of this intervention. The 12 checklist items are displayed in Table A. (available online), alongside how each has been interpreted for this intervention description.

The development process involved producing a 'pilot' manual that was used to standardise the delivery of the intervention in the feasibility study. The pilot manual is published alongside the feasibility study protocol[17] and available to view at http://www.trialsjournal.com/content/supplementary/1745-6215-14-109-s2.pdf.

During the feasibility study, feedback and other data were collected from participants and therapists, as per the study protocol,[17] and results relevant to the intervention development and description are presented in this article. This information was incorporated into the development of a revised 'final' intervention manual. To assist understanding of the intervention, the contents list and a selection of activity sheets taken from the final manual are provided in the 'Sample Section of Intervention Manual' online data supplement.

The entire final manual is not currently publicly available in order to minimise influence on usual care until a planned Phase III randomized controlled trial is completed. The final manual will then be made available in full, irrespective of outcome, on the 'Research Repository' of the University of the West of England at http://eprints.uwe.ac.uk.

1. *Name: Intervention title*

Home-based task-specific reach-to-grasp training after stroke.

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Table 1. Examples of reach-to-grasp activities for people with impaired shoulder and elbow movement.

Ability/functional level	Activity description(s)	Activity sheet
For people with significant shoulder flexion weakness.	The reaching task can be practised in side-lying, with the forearm resting on a bed so that the effect of gravity is minimised, thereby reducing the strength required to flex the shoulder.	31
For people who are unable to perform the whole action and need to improve elbow extension with shoulder flexion.	Task-specific part practice in sitting, e.g. with the hand resting on a table (to reduce the muscle force required as the limb is supported), shoulder flexion and elbow extension to touch a target object can be practised with/without grasp. This is then followed, when possible, by progressing to lifting the objects from the surface and varying direction/ extent of reach.	38 and 52
For people who can produce the whole action (e.g. able to open their hand and grasp an object), but may be slow or restricted in active range of shoulder flexion and/or elbow extension.	The whole reaching task can be practised in the sitting position, with variations such as the speed, extent of shoulder flexion or elbow extension or direction of the reach, for example: - varying direction of reach and incorporating grasp/release of objects of different sizes/shapes; - varying height/direction of reach and incorporating fine dexterity and precision	76 and 134

2. *Why: Rationale, definition and essential elements of the intervention*

To improve a functional skill, training needs to focus on practising that specific skill – a principle known as task-specific training. A variety of terms for task-specific training have been used in the literature.[5,7,18,19] The Cochrane review of repetitive task training defined studies eligible for inclusion as those '*where an active motor sequence was performed repetitively within a single training session, and where the practice was aimed towards a clear functional goal*'.[5] However, considering that many people with stroke cannot complete a task in its entirety, we expanded this definition to include 'part-practice'. 'Part-practice', as described by Carr and Shepherd,[20] involves undertaking tasks through segmentation; for example, the reach-to-grasp action is broken down into parts (e.g. shoulder flexion) that can be practised separately. Progression can take place by 'chaining', i.e. practising first one part, then practising that part with the next and so on, aiming for mastery of the whole task.[21] Consequently, for our intervention, 'task-specific reach-to-grasp training' was defined as: 'A *progressive training programme comprising practice of whole reach-to-grasp tasks and, where required, practice of the component parts that can be systematically reassembled into the whole task, with the aim of improving reach-to-grasp ability in daily activities*'.

The intervention was based on biomechanical analysis of functional reach-to-grasp movements and principles of motor learning.[22,23] To be 'task-specific', each action must specify an object to reach for, move towards or to grasp when either the whole task, or a component, is practised. Another requirement was that the practice should be carried out under conditions as similar to the conditions of the target task as possible to enhance carry-over into everyday living. Parameters considered were the: amplitude and direction of movement, available joint range, degrees of freedom, size/dimensions of the object(s), load, speed and the amount of gravity/friction to be overcome.

Examples and descriptions of reach-to-grasp activities, for people with impaired shoulder and elbow movement, are provided in Table 1. The

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activity sheets for these examples are provided in the 'Sample Section of Intervention Manual' online data supplement.

3. What – materials: Intervention manual

To enable standardised delivery of the intervention in the feasibility trial, the pilot manual of activities was designed based on the invariant kinematic features of reach-to-grasp actions, that is: shoulder flexion; scapular protraction and lateral rotation; shoulder external rotation; elbow extension; forearm supination; wrist extension and radial deviation; thumb abduction and opposition; finger extension with interphalangeal joints in some flexion.[22] Guidance notes for therapists based on the general principles of task-specific reach-to-grasp training prefaced the activities. These described procedures for the assessment of reach-to-grasp using a checklist of the invariant kinematic features of reach-to-grasp actions and how activities could be varied to suit the goals of each participant and optimise progression.

The pilot manual contained part-practice activities for each of the invariant kinematic features and a 'whole reach-to-grasp tasks' section, which incorporated sequential and simultaneous combinations of the invariant kinematic features into whole tasks. Activities for each part-practice section were based on the authors' clinical and research experience, along with suggestions from clinical colleagues and relevant exercises from an open-access physiotherapy website.[24] Part practice activities were ordered by difficulty (e.g. by the number of joints involved and the amount of gravity or resistance encountered). 'Whole task' activities covered the domains of self-care (e.g. brushing hair, putting on socks/shoes and tying laces, zipping/buttoning clothes), productivity (washing cutlery/crockery, using a knife and fork, pegging out and folding washing) and leisure (e.g. operating mobile phones and remote controls, knitting, playing cards or other games requiring dexterity), identified as common goals by people after stroke.[7,8]

All activities were discussed by the authors and included/excluded based on the previously agreed definition of task-specific reach-to-grasp training. Instructions were drafted for each activity and reviewed by the authors for clarity for participants. Photographs of the authors performing each activity were included. The pilot manual was reviewed by two people living with stroke, one with aphasia, who provided their views on the overall design and the feasibility of individual activities, following which the pilot manual was amended accordingly. The pilot manual, used in the feasibility study, contained 122 activities.

4. What – procedures: Role of therapist

At the initial intervention visit, the therapist delivering the intervention assessed each participant against a checklist of the invariant kinematic features to guide a biomechanical analysis of functional reach-to-grasp movements. Following this assessment, the therapist selected activities from the pilot manual and provided the participant with a folder containing colour copies of their activity sheets. At each subsequent visit, the therapist reassessed the reach-to-grasp movement and progressed/amended activities as appropriate for the individual. During each supervised session, the therapist aimed to maximise the number of repetitions of activities performed and encouraged participants to self-practise a maximum number of repetitions daily between visits.

5. Who provided: Therapist delivering intervention

The reach-to-grasp training was delivered primarily by one research physiotherapist with absence cover provided by two additional therapists (one a physiotherapist and one an occupational therapist) as required. All three therapists had worked as senior community therapists in the UK National Health Service and were trained and experienced in using upper limb task-specific training with people after stroke.

6. How: Method of delivery

The intervention was delivered on a one-to-one basis to promote engagement and motivation with

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the intervention and, importantly, allow specific feedback on performance; a key principle of skill acquisition. Extrinsic feedback (e.g. feedback provided by a therapist) can be focused either on the body movement (internal focus) or on the effects of the movement on the environment (external focus).[21] For this intervention, feedback on movement quality (i.e. an internal focus) was used only to avoid movement patterns that might potentially be detrimental to acquiring the required action; in all other instances, feedback was provided on the target/goal of the activity (i.e. an external focus). The choice of feedback focus was based on previous work on the role of feedback on improving reach-to-grasp function in people after stroke[25] and the findings of a recent systematic review that extrinsic feedback with an external focus, augments motor learning in stroke survivors.[26]

Family and carer engagement is strongly associated with good rehabilitation outcomes.[27,28] It is difficult however to standardise the potential influence and involvement of carers. Instructions for therapists delivering the reach-to-grasp intervention were therefore limited to advising carers to remind and encourage participants to practise, asking them to set up equipment and assist in recording the amount of practice.

7. Where: Environment and equipment

The intervention was delivered in participants' own homes, including care homes, as home-based rehabilitation has the potential to facilitate context-dependent learning; objects of relevance to the participant can be easily incorporated (e.g. comb, cutlery, cupboards) to allow occupational embedding of the activities.[29] As such, a variety of objects, preferably used by participants in everyday life, were selected for activity practice, as varying object shape and size can influence movement kinematics.[30]

8. When and how much: Timing and intensity of training

The intervention was designed to be delivered once participants had returned home from hospital and within the first 12 months after stroke. Participants ($n = 24$) who received the intervention in the feasibility study were a median of 111.5 days (interquartile range (IQR) 82.0, 241.0), approximately four months after stroke.

The intensity of the intervention encompasses the total amount of treatment, as well as its distribution over the treatment period in terms of the number, frequency and duration of individual treatment sessions, and the number of repetitions within each session. Uncertainty remains regarding optimal total therapy dosage for motor recovery after stroke, but there is evidence to support higher doses,[10,31] for example, the Extremity Constraint Induced Therapy Evaluation (EXCITE) trial of constraint-induced movement therapy demonstrated significant improvement in upper limb function after stroke with an intended dose of 60 hours of practice.[32] Consequently, with consideration given to the feasibility of delivery in community rehabilitation services, the treatment schedule for this reach-to-grasp intervention was set at a total target dose of 56 hours. This consisted of 14, one hour therapist visits over six weeks and additional self-monitored practice recommended for an hour a day, seven days a week. The frequency of therapist visits was tapered, three times a week in the first three weeks, twice in each of the next two weeks, then once in the final week, with the aim of increasing self-efficacy in practice and fostering self-management.

In terms of the number of repetitions within each session, high-intensity training was implemented. This is a key principle of the intervention based on animal studies of neuroplasticity[33–35] and meta-analyses of stroke rehabilitation trials.[10] However, no clear guidelines on the optimum number of repetitions currently exist.[28] A study of moderately affected stroke participants demonstrated that 300 repetitions were achievable within a one-hour training session.[36] With this in mind, and to include those with severe impairment, a target range of 100–300 repetitions/hour was endeavoured, dependent on individual participant's capabilities.

9. Tailoring: Individualising the intervention

The most important predictor of outcome for upper limb recovery is severity of motor impairment and function.[37] Participants receiving the intervention in

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this study had substantial loss of upper limb function: median Action Research Arm Test (ARAT) score = 8.5 (IQR 3.0; 24.0) (ARAT score range: minimum 0 – maximum 57).

Owing to the range of upper limb impairment, the therapist needed to select activities to suit the functional ability of each participant, with consideration also given to factors such as the patient's home environment, individual preferences, level of carer support, object shape and size, target positions and speed of movement.

As this was a structured intervention, the scope for individualised goal-setting was limited to each participant being encouraged to identify the tasks, objects and environments for practice, with the intention to make the intervention more personally relevant and stimulate engagement.[38]

10. How well – planned: How was adherence recorded

For the 14 treatment sessions, the number of repetitions and time spent on each activity from the pilot manual was recorded by the therapist using a tally counter and stop watch. For independent practice, participants were asked to record repetitions and time for each activity using log sheets covering the six-week duration of the intervention.

At the final intervention visit, a face-to-face, semi-structured interview was completed using a questionnaire designed to gauge the views of participants on the intensity and content of the intervention, as well as the design of the activity sheets. The questionnaire used is presented in the 'Intervention Questionnaire' online data supplement. The questions were asked by the research physiotherapist responsible for delivery of the intervention. While this may have biased responses, it was anticipated to facilitate a better understanding of any changes that should be made to the content of the intervention.

11. How well – actual: Results for intervention adherence and acceptability

A high intensity of therapist-supervised training was achieved, with 94% of planned visits completed (median 14; IQR 13.0, 14.0) and a median of 157 (IQR 96, 211) repetitions completed within each one-hour visit. Participants were engaged in upper limb practice for a median of 38.5 minutes (IQR 35.4, 48.8) per therapist-supervised session as time was required for assessment and to set-up and move between activities. During each one-hour therapist visit, the average number of activities practised was 4 (SD 1.3; range 1 to 7). Over the six-week intervention period, participants practised an average of 16.1 (SD 3.8; range 7 to 23) different activities. The majority of participants (17 of the 24) reported that practising independently for an hour per day was acceptable. Logs of participants' practice were poorly completed, but from those which were recorded, the median was only 30.0 minutes of independent practice per day (IQR 22.2, 44.5).

All 24 participants completed the treatment questionnaire. Analysis of responses shows that all participants who provided an answer ($n=23$) considered the intervention to be acceptable. A total of 16 participants considered the number of visits to be acceptable; the remaining eight would have preferred more. The intervention appeared to be provided at an appropriate level of difficulty with participants reporting a median of 6 out of 10 (IQR 5.0, 7.5) for how difficult the treatment was on a visual analogue scale where 0 represented extremely difficult and 10 represented extremely easy (Table 2).

Qualitative responses regarding the benefits of the treatment highlight participants' perceived improvements, particularly in confidence, e.g. 'more confidence to try things in my life' and 'made me feel more confident that things are going to move on in the future'. These translated into a wide variety of reported functional gains, such as being able to 'put on and take off wheelchair brake', 'put on own socks and tie shoe laces', 'steady a bowl/plate to eat with right hand', 'brush hair' and 'able to reach for Zimmer frame by myself, meaning I can walk by myself'. The full list of qualitative responses are presented in the accompanying 'Intervention Questionnaire' online data supplement.

The questionnaires also asked participants' opinions about the design of the activity sheets.

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Table 2. Responses to questionnaire about experience of reach-to-grasp intervention ($n = 24$). The questionnaire used and all qualitative responses (Questions 5 and 7) are reported in the online 'Intervention Questionnaire' data supplement.

		n	%	Median	IQR
1.	Number of visits acceptable	16	66.7%		
2.	One hour/day independent practice acceptable	17	70.8%		
3.	Reach-to-grasp intervention acceptable	23	95.8%		
4.	Reach-to-grasp intervention beneficial (0–10 VAS scale; 0 = not beneficial at all, 10 = extremely beneficial)			8.0	(7.5, 8.0)
6.	Reach-to-grasp intervention made difference to use of arm	20	83.3%		
8.	Reach-to-grasp intervention worth the amount of effort required	23	95.8%		
9.	Reach-to-grasp intervention difficult (0–10 VAS scale; 0 = extremely difficult, 10 = extremely easy)			6.0	(5.0, 7.5)

IQR: interquartile range; VAS: visual analogue scale.

The visual aspect of the activity sheets was evaluated positively by 16 of the 24 participants. However, two participants thought that the photographs would be improved by featuring people who have had a stroke rather than healthy individuals, commenting that '*it's better to have people who have had a stroke – see there are other people like you*' and that the current activity sheets show people who '*obviously don't have problems, which is a bit of a slap in the face*'. Three participants who had commented that the original photographs on the activity sheets were '*too straight-laced, not realistic*', '*could be more contemporary*' and '*should be "busier" like people's houses*'. Participants also commented that there should be more space for additional writing. Additionally, participants were asked which of the activities that they had practised they either 'liked' or 'disliked'.

12. Modifications: Revising the intervention following feasibility study

Following completion of the feasibility study, the pilot manual was revised, based on participant feedback and, following ethics committee approval and participant consent, to include photographs of participants with stroke practising the activities in their home environments. The authors then reviewed the ten activities that received more 'dislikes' than 'likes' from participants; it was decided that these activities had potential therapeutic merit and should continue to be included in the final manual, but the instructions and photographs be revised to improve understanding and acceptability.

The frequencies of activities prescribed to participants were calculated: 44 of the original 122 activities had not been assigned for independent practice. These 44 activities were individually reviewed by the authors and consensus decisions were made to remove ten of them. Next, a group consultation with the three therapists involved in the delivery of the intervention, one of whom was external to the research team and therefore able to provide a more independent assessment, was undertaken about their use of the pilot manual. The therapists were in favour of the number and variety of activities in the pilot manual, but reported difficulties finding and selecting activities from such an extensive resource. As a result, the final manual was re-structured to optimise ease of use with the classification of grip types described by Napier[39] incorporated into the indexing to structure the 'grasp' sections, while all sections were colour-coded to aid navigation. Additional activities suggested by the research therapists were included in the final manual, where authors were in agreement, with a final total of 144 activities. Changes to the format of the self-practice activity sheets were also made; the examples provided in the 'Sample Section of Intervention Manual' online data supplement are taken from the final manual.

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Discussion

This article has described, as per the TIDieR reporting guidelines,[16] an intensive task-specific reach-to-grasp training intervention for people after stroke for use in their own homes, as well as its development and piloting. Justification for the intervention is provided based on current best evidence and principles of skill acquisition. This process has resulted in a comprehensive intervention and final manual to support and structure its delivery. The intervention was considered acceptable by participants and therapists, and can be used in a subsequent pragmatic randomized controlled trial.

Standard study reports do not usually permit sufficient detail of intervention description, particularly for reports of complex non-pharmacological interventions.[12,13] The 'TIDieR' guidelines used here have provided a structured framework, which has ensured that all factors relevant to the delivery of the complex intervention have been described in detail. A limitation of the 'TIDieR' guidelines for this report is that they are primarily for describing an intervention, but not the process of how the intervention was developed. For this report therefore, a number of items required interpretation to also describe the rationale for decisions made during the development of the intervention.

Owing to the high-intensity of practice involved in the intervention, an important finding of the feasibility study was the acceptability of the intervention as evidenced by the completion of the majority of therapist visits and the completion of a median number of repetitions per visit within the pre-specified target range of 100–300 per hour. Within each visit, participants were engaged in upper limb practice for approximately two-thirds of the available therapy time, which interestingly replicates findings from a retrospective analysis of the EXCITE trial in which participants were engaged in upper limb training for 62% of the in-laboratory time.[40]

The majority of participants in this feasibility study considered an hour/day of independent practice to be acceptable. However, the self-report practice logs showed that participants actually practised, on average, for only half this time. This finding should be interpreted carefully as compliance with the logs was limited. Difficulties in accurately recording independent practice have been reported previously,[41] however methods for improving, as well as capturing, adherence in independent practice still need to be identified to improve the accuracy of trials and quality of clinical rehabilitation. To enhance intervention adherence, a number of practical strategies were identified and/or developed during the feasibility study.

- To encourage participants to focus on increasing repetitions rather than number of activities, it was established that a maximum of four activities would be provided for self-practise at any one time, i.e. additional activities needed to replace previous activities.
- To promote self-monitoring of repetitions during independent practice, participants were offered a hand-held tally counter. For some participants, this appeared to improve both adherence and accuracy of recording, although this requires further investigation.
- To achieve high numbers of repetitions, it was necessary for treatment sessions to be highly structured, which was perceived as different to usual care therapy sessions and some participants found challenging. Sessions appeared to be better tolerated, however, if they were described as 'training' rather than 'therapy' and an explanation provided on the intended effects of high-intensity training on neuroplasticity and motor learning.

Consultation with participants and therapists has led to an improved intervention manual intended to be used to structure and standardise the delivery of the training intervention in a randomized controlled trial. Although this manual is currently designed to be used in print format, an electronic version could be used in the future, for example in the context of telemedicine, where therapists could provide feedback and suggest new activities, and patients could report their experiences, and photographs/videos of recommended exercises or patient performance could be uploaded. The current study is an initial step in the evolution of the content and delivery of the intervention.

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Clinical messages

- A home-based task-specific reach-to-grasp training programme for people after stroke has been described and a manual has been developed to enable systematic and standardised delivery.
- The high-intensity task-specific intervention was considered acceptable and beneficial by participants and feasible for use in the home setting.

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Contributors

Paul Cunningham developed the intervention and manual, testing feasibility, writing – was lead author. Dr Ailie Turton developed the intervention and manual, testing feasibility, writing – contributed significant text. Professor Frederike vanWijck and Professor Paulette VanVliet developed the intervention and manual, writing – contributed significant text.

Declaration of Conflicting Interests

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